SIMATS ENGINEERING
ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

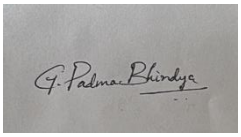
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I,**G. Padma Bhindya (192565049)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message.
Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
Demonstrate how the encryption process transforms the message step-by-step.
Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
Analyze how investigators could detect the presence of hidden data within the image.
Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

ANSWERS

SCENARIO 5: LEGACY ENCRYPTION IN CORPORATE COMMUNICATION

Introduction:

In this scenario, a company is using a legacy **transposition-based encryption method with three levels of arrangement**. The received ciphertext is:

WECRLTEERDSOEEFEAOCAIVDEN

In a transposition cipher, the letters themselves are not changed. Instead, their positions are rearranged according to a particular pattern. Here, the three-level arrangement can be reversed using a **3-rail zigzag pattern**.

a) Determine the Most Probable Original Message

Step 1: Given Ciphertext

Ciphertext:

WECRLTEERDSOEEFEAOCAIVDEN

There are **24 letters** in the ciphertext.

Step 2: Divide the Ciphertext into Three Rails

For a 3-level arrangement, the ciphertext is divided into three rows according to the number of characters that occur on each rail.

Rail Ciphertext Characters

Rail 1 W E C R L T E

Rail 2 E R D S O E E F E A O C

Rail 3 A I V D E N

Therefore:

Rail 1 → W E C R L T E

Rail 2 → E R D S O E E F E A O C

Rail 3 → A I V D E N

Step 3: Arrange the Rails in a Zigzag Pattern

For three rails, the reading pattern is:

Rail 1 → Rail 2 → Rail 3 → Rail 2 → Rail 1 → Rail 2 → Rail 3 → ...

The characters are therefore selected from the three rows in this order.

Clear Decryption Table

Position	Rail Used	Character	Reconstructed Text
1	Rail 1	W	W

Position	Rail Used	Character	Reconstructed Text
2	Rail 2	E	WE
3	Rail 3	A	WEA
4	Rail 2	R	WEAR
5	Rail 1	E	WEARE
6	Rail 2	D	WEARED
7	Rail 3	I	WEAREDI
8	Rail 2	S	WEAREDIS
9	Rail 1	C	WEAREDISC
10	Rail 2	O	WEAREDISCO
11	Rail 3	V	WEAREDISCOV
12	Rail 2	E	WEAREDISCOVE
13	Rail 1	R	WEAREDISCOVER
14	Rail 2	E	WEAREDISCOVERE
15	Rail 3	D	WEAREDISCOVERED
16	Rail 2	F	WEAREDISCOVEREDF
17	Rail 1	L	WEAREDISCOVEREDFL
18	Rail 2	E	WEAREDISCOVEREDFLE
19	Rail 3	E	WEAREDISCOVEREDFLEE
20	Rail 2	A	WEAREDISCOVEREDFLEEA
21	Rail 1	T	WEAREDISCOVEREDFLEEAT
22	Rail 2	O	WEAREDISCOVEREDFLEEATO
23	Rail 3	N	WEAREDISCOVEREDFLEEATON
24	Rail 2	C	WEAREDISCOVEREDFLEEATONC
25	Rail 1	E	WEAREDISCOVEREDFLEEATONCE

Decryption Result

The reconstructed plaintext is:

WEAREDISCOVEREDFLEEATONCE

Adding spaces and punctuation gives:

“WE ARE DISCOVERED. FLEE AT ONCE.”

Thus, the most probable original message is:

WE ARE DISCOVERED. FLEE AT ONCE.

b) Why the Method Is Vulnerable in Modern Communication Systems

The three-level transposition method is vulnerable because it only **rearranges the positions of characters** rather than providing strong modern encryption.

1. The letters remain unchanged

The original letters are still present in the ciphertext. Only their order is changed. This provides an attacker with useful information for cryptanalysis.

2. Limited arrangement

The three-level pattern is relatively simple and predictable. If an attacker identifies the type of transposition used, they can attempt to reconstruct the original arrangement.

3. Easy to attack with modern computers

Modern computers can quickly test possible arrangements and patterns. Therefore, a simple classical transposition method does not provide adequate protection for confidential corporate information.

4. Weak confidentiality

The method does not provide the strong mathematical security offered by modern encryption algorithms. Sensitive corporate messages could potentially be recovered if the encryption pattern is discovered.

5. No proper integrity protection

A basic transposition cipher does not provide a mechanism to reliably detect whether an encrypted message has been modified during transmission.

Therefore, this legacy method is unsuitable for protecting sensitive information in modern communication systems.

c) Practical Enhancement

A practical enhancement would be to replace the legacy transposition cipher with a modern authenticated encryption algorithm such as **AES-GCM**.

AES-GCM provides:

- **Confidentiality:** prevents unauthorized users from reading the message.
- **Strong encryption:** makes unauthorized decryption computationally difficult.
- **Integrity:** helps detect whether the encrypted message has been modified.
- **Authentication:** allows the receiver to verify the authenticity of the encrypted data.

The company should also implement **secure cryptographic key management**, including secure generation, storage, distribution, and regular management of encryption keys.

Conclusion

The ciphertext **WECRLTEERDSOEEFEAOCAIVDEN** can be decrypted using the three-level

zigzag transposition arrangement. The resulting plaintext is:

"WE ARE DISCOVERED. FLEE AT ONCE."

Although the transposition method demonstrates the basic concept of classical encryption, it is vulnerable because it only rearranges characters and provides limited security. Replacing it with **AES-GCM** and implementing proper key management would provide much stronger confidentiality and integrity for corporate communication.

SCENARIO 1: EVENT COMMUNICATION SECURITY

Introduction:

In this scenario, a student organization wants to securely share confidential event details with its core members. A simple classical encryption technique using a **fixed shift pattern** is applied to the message. The given message is **"MEET AT NOON"**, and each letter is shifted by **4 positions** in the alphabet. This technique is known as the **Caesar Cipher**.

1. Encryption Process

The Caesar Cipher replaces each letter in the original message with another letter that is a fixed number of positions away in the alphabet. Here, the shift value is **4**, so every letter is moved **four positions forward**.

For example:

- M → N → O → P → **Q**
- E → F → G → H → **I**
- T → U → V → W → **X**
- A → B → C → D → **E**
- N → O → P → Q → **R**
- O → P → Q → R → **S**

Applying this shift to the complete message:

Original Message:

MEET AT NOON

Encrypted Message:

QIIX EX RSSR

The spaces between the words are retained to make the encrypted message easier to read.

2. Encryption Table

Original Letter	**M**	**E**	**E**	**T**	**A**	**T**	**N**	**O**	**O**	**N**
Shift	+4	+4	+4	+4	+4	+4	+4	+4	+4	+4

Original Letter	**M**	**E**	**E**	**T**	**A**	**T**	**N**	**O**	**O**	**N**
Encrypted Letter	Q	I	I	X	E	X	R	S	S	R

Therefore, the encrypted form of **"MEET AT NOON"** is:

QIIX EX RSSR

3. Decryption Process

An authorized member who knows the encryption rule can recover the original message easily. Since encryption was performed by shifting every letter **4 positions forward**, decryption is performed by shifting every letter **4 positions backward**.

For example:

- Q → P → O → N → **M**
- I → H → G → F → **E**
- X → W → V → U → **T**
- E → D → C → B → **A**
- R → Q → P → O → **N**
- S → R → Q → P → **O**

Thus:

Encrypted Message:

QIIX EX RSSR

After shifting 4 positions backward:

MEET AT NOON

4. Role of the Encoding Rule

The shift value of **4** acts as the encoding rule or key for this communication. Only a member who knows this rule can directly reverse the transformation and understand the original message. The same rule must be applied consistently to all letters during encryption and its reverse during decryption.

Conclusion

The given message **"MEET AT NOON"** is encrypted using a Caesar Cipher with a fixed shift of **4 positions**, producing **"QIIX EX RSSR."** An authorized member can retrieve the original message by shifting every encrypted letter **4 positions backward**. This demonstrates the basic working principle of classical substitution encryption, where the original characters are replaced according to a fixed rule.